

Decentralized Markets for Crowdsourced Mathematics

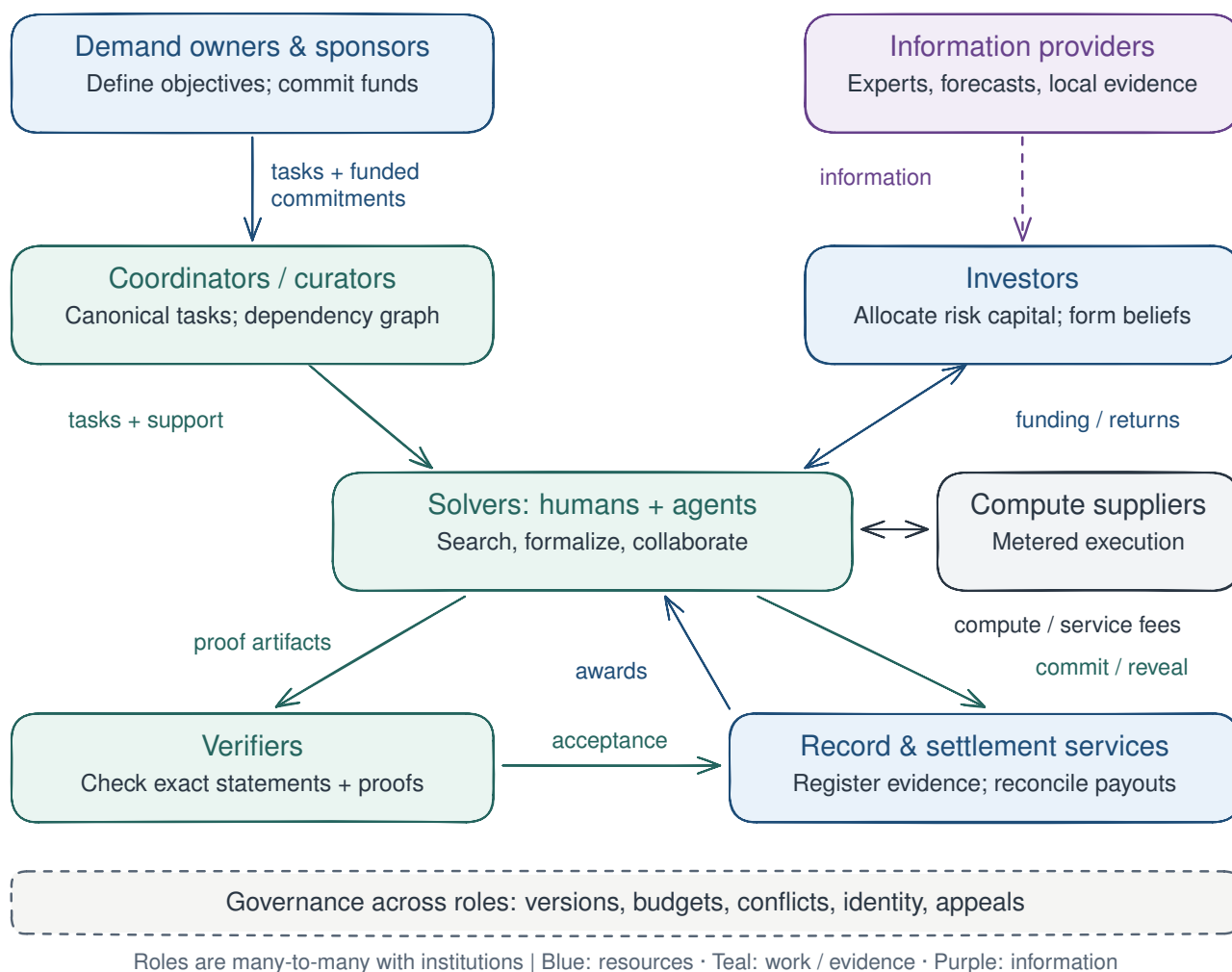
Roles, Incentives, and Institutional Design

Varifold Lab

1. What this paper does

AI expands the scale of proof search, while formal proof assistants make accepted results independently checkable. The economic problem is how to fund uncertain attempts and coordinate contributions before a proof exists.

Our contribution. We develop an abstract framework connecting demand, funding, proof work, verification, and settlement. We compare support for exploration, intermediate rewards, and competing submissions; analyze investment under heterogeneous beliefs and coordination across platforms; and relate these choices to existing initiatives and testable research questions.



A functional framework. Each box denotes a role. Several institutions can perform the same role, and one institution can perform several roles. Arrows show principal flows of resources, work, and information.

Reading the map. Sponsors value outcomes; investors finance work for contingent returns; solvers perform it. Coordination defines tasks, verification checks proofs, and settlement applies reward rules. Separating these roles makes responsibilities and conflicts visible across institutional arrangements.

Scope. The contribution is conceptual: a framework and mechanism comparisons whose effects require empirical evaluation. It accommodates different degrees of decentralization, conventional payments, and crypto-based settlement.

2. Funding choices and investment decisions

Exploration funding

We compare two ways to finance eligible attempts, including those that fail. **Advance compute credits** reduce liquidity barriers and cap exposure, but restrict provider choice and can encourage exhausting allowances. **Capped reimbursement** offers more execution choice, but requires participants to front the money and bear audit risk. Both require a finite, committed budget and controls against duplicate claims.

The audit problem has three layers: actual consumption, task relevance, and productive effort. Receipts and task-bound artifacts support the first two; productivity remains harder to establish. This creates a tradeoff between supporting exploration and the cost of monitoring it. A subsidy lowers private cost while the underlying computation remains a real resource cost.

Intermediate rewards and cumulative progress

We examine rewards for reviewed supporting lemmas before a root theorem is complete. A completed lemma can earn its award even if another branch later fails, enabling specialized contributors to participate. Conditional reductions remain conditional until their dependencies are closed. The main tradeoff is between financing useful partial progress and encouraging arbitrary fragmentation. Reviewed task admission, separate integration rewards, and funded reuse awards address different parts of this problem. Scholarly credit and financial claims require distinct rules.

Priority and shared rewards

We compare exclusive priority with sharing a fixed terminal pool among eligible submissions committed within an announced window. A common commitment phase followed by revelation limits copying from protocol disclosures. Sharing can reduce sensitivity to small timing differences and recognize parallel work, but can dilute incentives and invite duplicate or Sybil claims. The window, disclosure schedule, and unit of entitlement therefore matter. Recorded priority supplies evidence for a payment rule; it does not establish independent discovery.

Investment under heterogeneous beliefs

We express investor i 's assessment of action a using its own information $\mathcal{I}_{i,t}$ and subjective beliefs $Q_{i,t}$:

$$\Pi_{i,t}(a) = \mathbb{E}_{Q_{i,t}}[R_i(a) - C_i^{\text{net}}(a) \mid \mathcal{I}_{i,t}].$$

Here R_i is the investor's contracted receipts and C_i^{net} its costs after eligible support. Expert advice, prediction prices, and local experiments inform this private evaluation. We distinguish belief in a theorem's truth, success within budget, award eligibility, and eventual profitability. Competition changes both the chance of a payout and its size; liquidity, risk preferences, and the option to stop also affect investment.

Implication across the comparisons. Sponsor objectives and reward rules shape which work is privately attractive. Evaluating a mechanism requires measuring both participant returns and the scientific benefits produced per unit of total resources.

3. Institutional practice and research priorities

How existing initiatives inform the framework

We relate documented practices to the roles above and identify the additional arrangements needed to connect them. These initiatives illustrate parts of the framework; the proposed combinations remain to be evaluated.

Demand and sponsorship. DARPA expMath and AWS’s provable-security work illustrate scientific and corporate demand. The Justin Sun Prize combines crypto-denominated rewards with adjudication. These examples help distinguish valuing an outcome, financing its production, and deciding when an award is due.

Coordination and settlement. OpenMath’s tooling describes theorem discovery, local proving, commitments, revelation, and reward claims. Prove2Me organizes missions, canonical statements, proof-sketch dependencies, and contribution credit. Their workflows provide settings for examining task identity, intermediate rewards, and verification delays. Funding those extensions requires explicit budgets and financial entitlements.

Information. Kalshi’s filed Millennium contract concerns an official prize-award event. Expert assessments and local proof experiments can also inform investors’ beliefs. Using these inputs in a private investment decision requires matching each assessment to the investor’s task, budget, and horizon.

Cross-platform coordination and institutional risks

We identify common task versions, evidence bundles, and reconciled payment claims as requirements for coordination across platforms. A valid proof, an eligible contribution, and a payable entitlement are separate determinations. Shared records support these decisions alongside rules for authority, conflicts, and appeals.

Design problem	Central tradeoff
Formal targets	Mechanical acceptance supports clear contracts; statement meaning, assumptions, and scientific value still require review.
Effort and attribution	Audits and eligibility checks can deter waste and duplicate claims, while raising privacy, entry, and administration costs.
Authority and information	Shared services can improve coordination, while concentrated providers and financially interested participants create correlated risks.
Funding credibility	Escrow and reconciled claims support payment commitments; reliable service capacity and solvency require separate evidence.

The empirical agenda

Funding: With the same total budget, which mix of exploration support, lemma rewards, and terminal prizes produces more accepted and reused work?

Sharing: How do reward windows and disclosure schedules affect completion, duplicate effort, entry, and disputed payments?

Information: Do forecasts improve allocation compared with simple baselines when unsuccessful and censored attempts are included?

Across these tests, the outcome measures are mathematical progress, total compute and review cost, participation, and credible settlement. Institutional experience can refine both the assumptions and the mechanisms being compared.

Companion manuscript, Version 2. Sections 2–5 develop the framework and mechanisms; Section 6 discusses institutional examples; Section 7 develops the research agenda. Full references and source qualifications are provided there.